LESSON PLAN

ADVANCED AIRCRAFT TECHNOLOGY (20)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	10
Total No. of Lesson Plans	20	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction to Special purpose oscilloscope (Digital storage oscilloscope and sampling CRO)
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction to Special purpose oscilloscope (Digital storage oscilloscope and sampling CRO)
5. Specific Objectives	20.3.1.1 Trainee will be able to explain the construction and operation of DMM & CRO.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Demonstration of DSO and Sampling CRO block diagrams; Safety: High voltage precautions; Supervision: Instructor monitors connections.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a standard CRO?
(b) Define the term 'sampling' in the context of signal acquisition.
(c) List two limitations of an analog CRO regarding signal storage.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Digital Storage Oscilloscope (DSO) Architecture • Analog-to-Digital Conversion Process • Memory Depth and Bandwidth Specifications • Waveform Capture and Triggering Mechanisms • Display Resolution and Update Rates	L & Ds	PPT, IB	15 min
(b)	• Sampling Oscilloscope Principles • Real-Time vs Equivalent Sampling • Nyquist Theorem and Aliasing Effects • Sample Rate Requirements for High Frequency • Reconstruction of Continuous Waveforms	L & Ds	PPT, PDF	20 min
(c)	• Comparison: DSO vs Sampling CRO • Application Scenarios for Special Purpose Units • Advantages of Digital Storage Technology • Limitations of Sampling Techniques • Selection Criteria for Aircraft Maintenance	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how a DSO converts an analog input signal into a digital display.
(b) Differentiate between real-time sampling and equivalent sampling methods.
(c) Describe a scenario where a sampling CRO is preferred over a real-time DSO.

12. Home Assignment and Follow Up	Prepare a brief note comparing the architecture of a standard CRO with a Digital Storage Oscilloscope, highlighting the role of the ADC in the latter, to be discussed in LP 11 regarding Glass Cockpit architecture.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Concept of Glass cockpit and architecture, Ring Laser Gyros. Practical demonstration of Gyro properties like rigidity and precision.	Refer to the master syllabus reference list	LP 11

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min